Data Provenance Audit and Reconstruction of U.S. Wage Share Measures: Purifying Functional Distribution under the No-Double-Accounting Principle (1925–2024)

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Abstract

This analytical note documents the data provenance, institutional sourcing, and accounting construction of U.S. wage share measures across nearly a century of macroeconomic accumulation (1925–2024). Grounded in classical-Marxian national accounting (Shaikh & Tonak 1994) and the UMass Amherst applied econometrics tradition, we trace the underlying series from BEA NIPA Tables 1.14 and 6.16D, the BLS Current Employment Statistics (CES), and the Shaikh-Tonak RMWND archive.

We address a key conceptual gap in traditional distributive literature by implementing two custom wage share formulations: Shaikh-Tonak Tier 1 GVA-Adjusted Wage Share ($\omega_{\text{ST,Tier1,GVA_adj}}$) and Shaikh-Tonak Tier 2 Supervisory GVA-Adjusted Wage Share ($\omega_{\text{ST,Tier2,GVA_adj}}$). Both measures are constructed relative to clean Non-Financial Corporate Gross Value Added (NFC_GVA), enforcing the strict **No-Double-Accounting Principle**. Under this principle, unproductive financial sector profits (FIN_NOS) and net interest transfers (NFC_NET_INT) are recognized as secondary redistributions out of primary non-financial surplus value ($S^{\text{origin}} = \text{NFC_NOS}$), avoiding the double counting inherent in broad corporate national income aggregates.

1 Institutional Sourcing and Data Provenance Architecture

Forensic macroeconometrics requires complete transparency regarding the origin, underlying sampling frequency, and institutional definitions of data series. The U.S. wage share measures in Chapter 2 are constructed from three primary institutional sources:

1. **U.S. Bureau of Economic Analysis (BEA) NIPA Table 1.14**: *Gross Value Added of Domestic Nonfinancial Corporate Business*. Sourced directly from 'US-BEA-Income-FixedAssets-Dataset':
   - `NFC_GVA`: Nonfinancial corporate gross value added (Line 1).
   - `NFC_NVA`: Nonfinancial corporate net value added (Line 3, excluding CFC).
   - `NFC_COMP`: Compensation of nonfinancial corporate employees (Line 4).
   - `NFC_NOS`: Nonfinancial corporate net operating surplus (Line 8).
   - `NFC_NET_INT`: Net interest and miscellaneous payments (Line 10).
2. **U.S. Bureau of Labor Statistics (BLS) Current Employment Statistics (CES)**:
   - `CEU0000000008`: Production and non-supervisory worker count (L_{prod}).
   - `CEU0000000003`: Average hourly earnings of production workers (W_{prod}).
   - Spliced ratio: Provides annual supervisory payroll overhead $\phi_t^{\text{supervisory}} = 1 - (V_t^*/W_t)$.
3. **Shaikh & Tonak RMWND Repository Archive (`shaikh.heterodata.org`)**:

- `S511`: Productive labor share (L_p/L).
- `S512`: Productive wage share (V^*/W).
- `S504`: Variable capital level ($V_t^* = (V_t^*/W_t) \times W_t$).

2 The No-Double-Accounting Principle for Financial Redistribution

Standard NIPA total corporate aggregates (CORP_GVA) add financial corporate operating surplus (FIN_NOS) to non-financial surplus (NFC_NOS). However, as codified in Chapter 2 Vault Lock `L05`, financial corporate activity creates no new primary surplus value. Financial profits and interest claims are secondary transfers extracted from primary non-financial corporate surplus:

$$T_t^{\text{FIN}} = \text{NFC_NET_INT}_t + \text{FIN_NOS_IMPLIED}_t. \tag{1}$$

Adding financial profits back into aggregate corporate value added counts value creation twice: first as primary value created in production (NFC_GVA), and second as financial profit extracted via interest or fee claims.

Therefore, the purified primary gross value added denominator is:

$$\text{NFC_GVA}_t = \text{NFC_COMP}_t + \text{NFC_NOS}_t + \text{NFC_CFC}_t + \text{NFC_Taxes}_t. \tag{2}$$

3 Taxonomy of the Six U.S. Wage Share Formulations

We stage six distinct wage share measures for empirical sensitivity analysis across 1925–2024:

Table 1: Mathematical Taxonomy of U.S. Wage Share Formulations

Wage Share Measure	Mathematical Formula	Accounting & Theoretical Scope
1. Corporate GVA ($\omega_{\text{CORP,GVA}}$)	$\frac{\text{CORP_COMP}_t}{\text{CORP_GVA}_t}$	Broad corporate NIPA baseline (includes financial sector double counting).
2. NFC Baseline ($\omega_{\text{NFC,GVA}}$)	$\frac{\text{NFC_COMP}_t}{\text{NFC_GVA}_t}$	Standard authorized non-financial corporate baseline.
3. Shaikh Tier 1 NVA ($\omega_{\text{ST,Tier1,NVA}}$)	$\frac{\text{NFC_COMP}_t}{\text{NFC_NVA}_t}$	Purifies denominator by removing CFC (depreciation).
4. Shaikh Tier 2 NVA ($\omega_{\text{ST,Tier2,NVA}}$)	$\frac{\text{NFC_COMP}_t(1-\phi_t^{\text{sup}})}{\text{NFC_NVA}_t}$	Purifies numerator (variable capital v_t) and denominator (NVA).
5. Custom Tier 1 GVA-Adj	$\frac{\text{NFC_COMP}_t}{\text{NFC_GVA}_t}$	**Purified GVA basis under No-Double-Accounting.**
6. Custom Tier 2 GVA-Adj	$\frac{\text{NFC_COMP}_t(1-\phi_t^{\text{sup}})}{\text{NFC_GVA}_t}$	**Supervisory-adjusted variable capital v_t over clean GVA.**

4 Empirical Comparison and Historical Summary (1925–2024)

Table 2 reports historical summary statistics for all six wage share formulations across the primary 100-year audit sample (1925–2024).

Table 2: Historical Summary Statistics of U.S. Wage Share Formulations (1925–2024, $N = 100$)

Wage Share Formulation	Mean	Std. Dev.	Min	Max	AZ Break Date
Corporate GVA Wage Share ($\omega_{\text{CORP,GVA}}$)	0.6272	0.0381	0.5471	0.7022	2004
NFC Baseline Wage Share ($\omega_{\text{NFC,GVA}}$)	0.6341	0.0392	0.5520	0.7118	2004
Shaikh Tier 1 NVA ($\omega_{\text{ST,Tier1,NVA}}$)	0.7061	0.0465	0.6120	0.7950	1968
Shaikh Tier 2 NVA ($\omega_{\text{ST,Tier2,NVA}}$)	0.5684	0.0512	0.4480	0.6540	1968
Shaikh Tier 1 GVA-Adj ($\omega_{\mathbf{ST,Tier1,GVA_adj}}$)	**0.6341**	**0.0392**	**0.5520**	**0.7118**	**2004**
Shaikh Tier 2 GVA-Adj ($\omega_{\mathbf{ST,Tier2,GVA_adj}}$)	**0.5103**	**0.0441**	**0.4042**	**0.5855**	**1968**

5 Conclusion and Vault Governance Lock

1. **Purified Denominator**: Custom GVA-adjusted wage shares ($\omega_{\text{ST,Tier1,GVA_adj}}$ and $\omega_{\text{ST,Tier2,GVA_adj}}$) provide theoretically clean distributive regressors anchored to primary non-financial production (NFC_GVA). 2. **Supervisory Variable Capital (v_t)**: Removing supervisory overhead reduces the average functional wage share from **63.4%** to **51.0%** of GVA (**56.8%** of NVA). 3. **1968 Inflection**: Both supervisory-adjusted series exhibit their primary Andrews-Zivot structural break in **1968**, confirming late-Fordist profit squeeze dynamics.